

d) The absolute value of the sum of two integers does not exceed the sum of the absolute values of these integers.

$$\forall x \forall y (|x+y| \leq |x| + |y|)$$

21) Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that every positive integer is the sum of the squares of four integers.

$$\forall n (n > 0) \Rightarrow \exists a \exists b \exists c \exists d (n = a^2 + b^2 + c^2 + d^2)$$

22) Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.

$$\exists x ((x > 0) \Rightarrow \forall y \forall z \forall w (x \neq y^2 + z^2 + w^2))$$

23) Express each of these mathematical statements using predicates, quantifiers, logical connectives and mathematical operators.

Domain: The set of all real numbers

a) The product of two negative real numbers is positive

$$\forall x \forall y ((x < 0) \wedge (y < 0) \Rightarrow (xy > 0))$$

b) The difference of a real number and itself is zero.

$$\forall x (x - x = 0)$$

c) Every positive real number has exactly two square roots

$$\forall n ((n > 0) \Rightarrow \exists y \exists z ((y \neq z) \wedge (y^2 = n) \wedge (z^2 = n) \wedge \forall w ((w^2 = n) \Rightarrow (w = y \vee w = z))))$$

d) A negative real number does not have a square root that is a real number.

$$\forall x ((x < 0) \Rightarrow \forall y (y^2 \neq x))$$

24) Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.